PREDICTING OCCUPANCY EFFIENCY IN SEATLLE AIRBNB LISTINGS

airbnb

COURSE : IST5520 – Data Science And Machine Learning With Python

TEAM NAME : Krypton Squad

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DATE: 11/20/2025

WHY THIS PROJECT MATTERS?

- Most studies focus on price prediction or yearly revenue, but:
  - Hosts struggle more with vacancy than with setting price.
- Occupancy drives revenue:
  - Revenue ≈ Price × Nights Booked × Occupancy Rate
- Our project helps:
  - New hosts: understand what drives early bookings.
  - Existing hosts: identify levers to move from low to high occupancy.
- We focus on a 30-day window:
  - Short, actionable horizon.
  - Hosts can quickly test and adjust their strategy.

DATA OVERVIEW

- Dataset: Seattle Airbnb dataset, Washington, United States (Lisiting.xlsx + Review.csv)
- Size: 4,469 listings
- Original Listing data : (6862, 80)
- After adding two new columns (1. Estimated_revenue_30 & 2. Estimated_occupancy_30) : (6862, 82)
- Final Features (18 columns) : property_type, room_type, neighbourhood_cleansed, latitude, longitude, accommodates, price, estimated_occupancy_30, estimated_revenue_30, availability_30, availability_365, instant_bookable, host_since, host_is_superhost, host_response_rate, host_total_listings_count, number_of_reviews, review_scores_rating.
- Source: Inside Airbnb: https://insideairbnb.com/get-the-data/

Why We're Presenting Again

Initial issue: We used 30-day data but analyzed yearly revenue.

Revision: Shifted to a correct target — 30-day occupancy.

Upgrade: Added new features + cleaned metrics + merged reviews.

Result: A clearer, more accurate, and consistent occupancy model.

GOAL - FROM WRONG METRIC TO BETTER GOAL: FIXING OUR FIRST ATTEMPT

New

Old Goal: Nov 12 Presentation

Target : Yearly revenue (estimated_revenu_365)

Problem :

Used 30-day based data but jumped to annual revenue.

Mixed estimated_revenue_365 with 30-day variables.

Professor’s feedback: Mismatch between time window and target.

New Goal: This Presentation

Target : 30-Day Occupancy Efficiency

occupancy_rate_30 = estimated_occupancy_30 / 30

Problem :

Keeps time window consistent (30 days everywhere).

Directly measures how booked a listing is.

Hosts can change their behavior to influence it.

DATA CLEANING HIGHLIGHTS

- Converted messy strings (price, response rate) into numeric.
- Imputed 0 or missing values.
- estimated_occupancy_30 using 30 - availability_30 when occupancy was 0 but listing was available.
- estimated_revenue_30 = price × estimated_occupancy_30 when 0.
- Removed unrealistic values and enforced logical rules.

KEY FEATURES WE SELECTED & WHY THEY MATTER FOR OCCUPANCY

Category	Features	Why it Matters
1. Pricing & Revenue Behavior	price, log_priceestimated_revenue_30, log_estimated_revenue_30	balance between being affordable vs. premium; too high price may reduce bookings, but right price for high demand listings increases occupancy.
2. Availability & Booking Window	availability_30, availability_365	very high availability may indicate low bookings; patterns here signal demand vs. oversupply.
3. Host & Listing Quality	host_is_superhost, instant_bookablehost_response_rate, host_years_activenumber_of_reviews, log_reviewsreview_scores_rating	trust + convenience → guests more likely to book.
4. Capacity & Property Type	accommodatesproperty_type, room_type	different guest segments (families vs solo travelers) → different demand patterns.
5. Location	dist_km_downtownneighbourhood_cleansed	distance to downtown + popular neighborhoods strongly affect occupancy.

REDUCING MULTICOLLINEARITY WITH VIF

Many features are highly correlated (e.g., estimated_occupancy_30 and occupancy_rate_30, price and log price, etc.). It Makes regression unstable and harder to interpret.

So, What we did ?

Computed Variance Inflation Factor (VIF) for numeric predictors.

Iteratively:

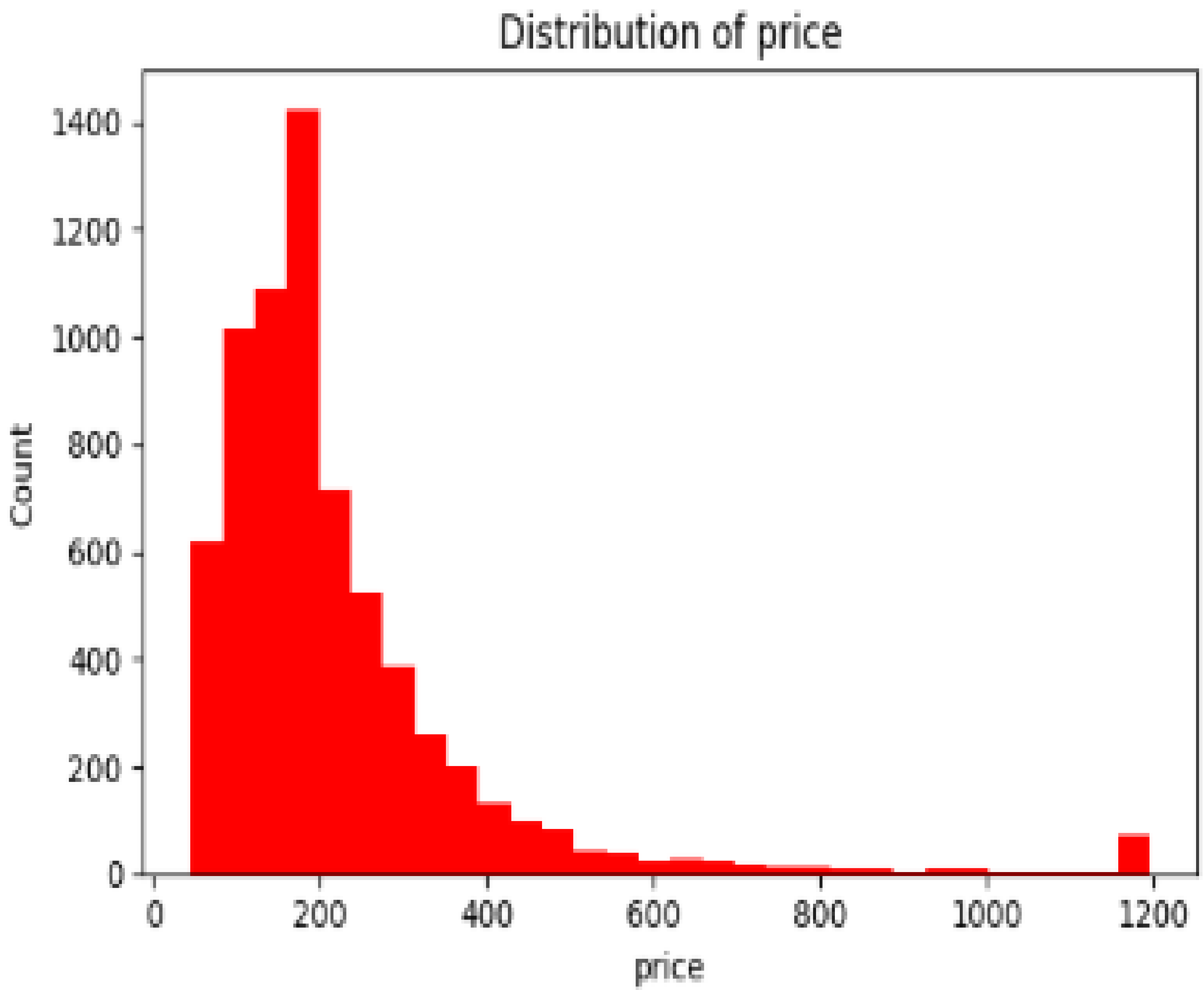
- Dropped the feature with highest VIF > 10.
- Recalculated VIF.
- Stopped when all remaining features had acceptable VIF.

EDA FINDINGS & OUTLIER TREATMENT

Price

Skewed right (few very expensive listings).

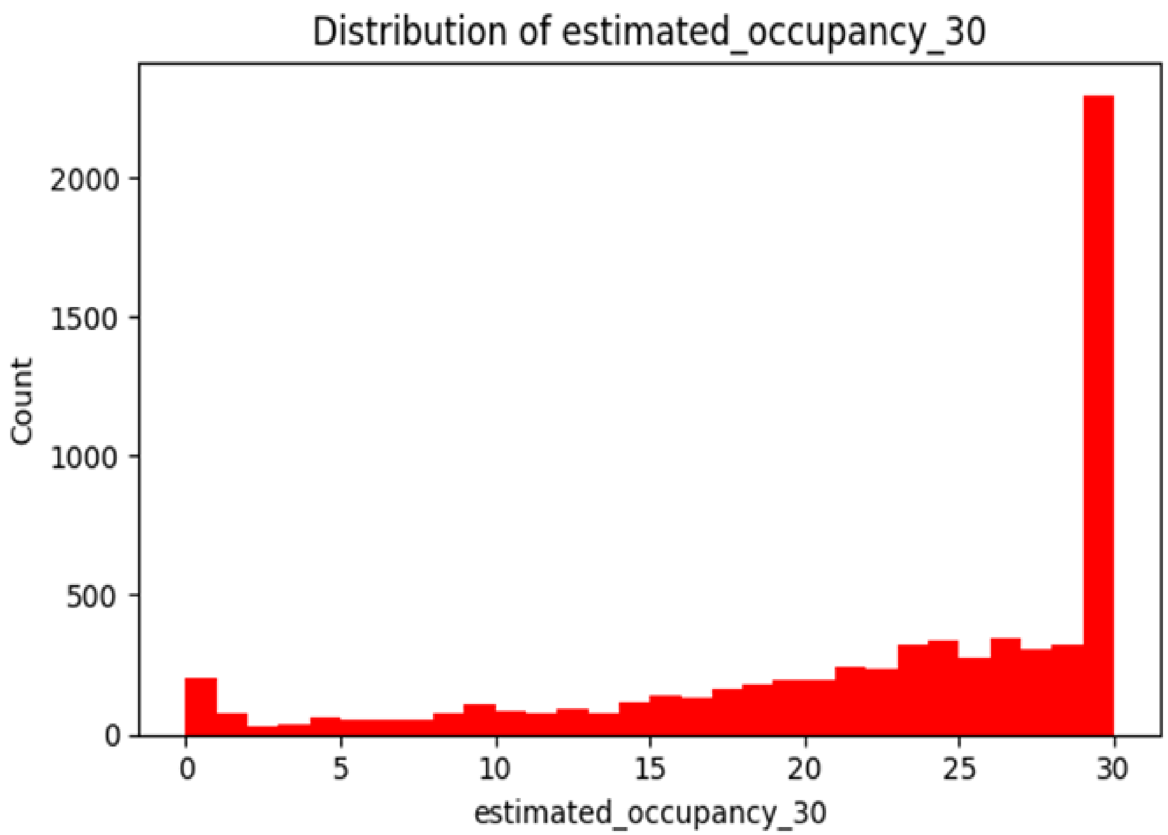
We clipped extreme 1% high/low values.



EDA FINDINGS & OUTLIER TREATMENT

Estimated Occupancy

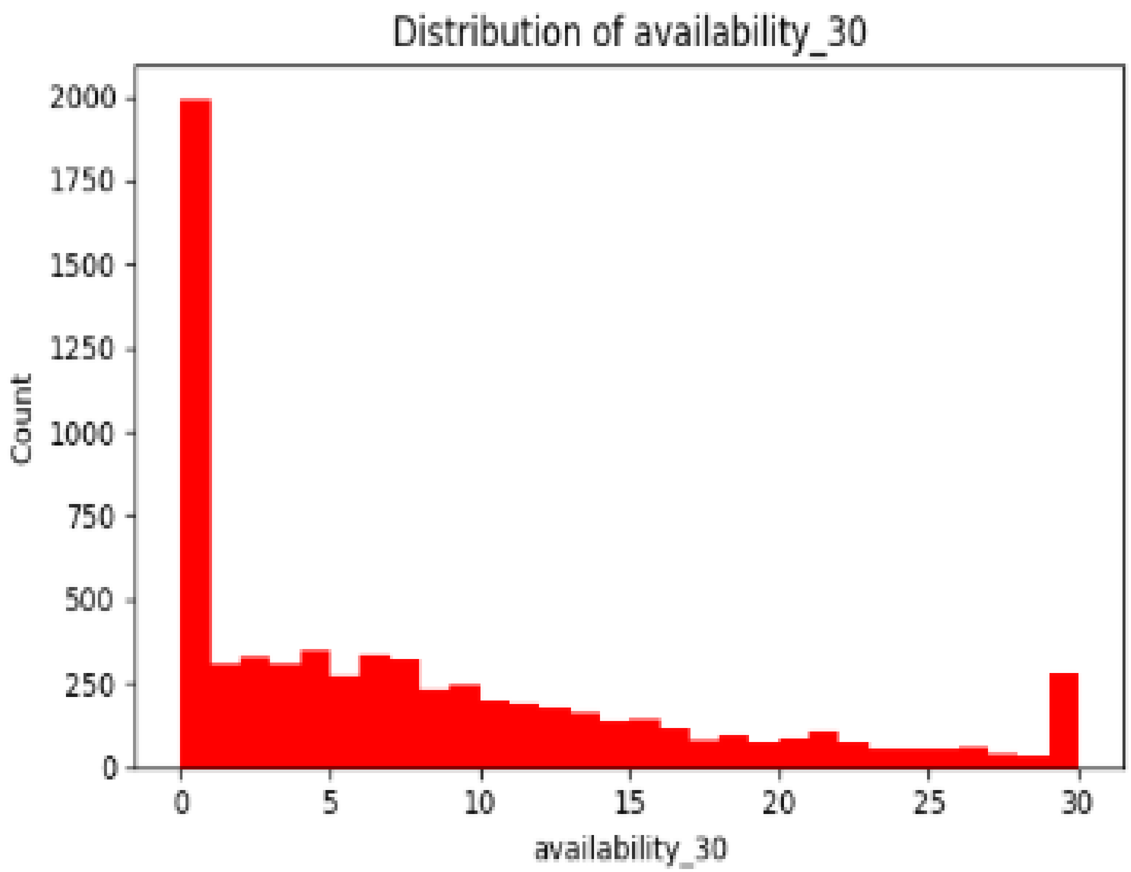
Many listings are moderately occupied, some very low, some very high.



EDA FINDINGS & OUTLIER TREATMENT

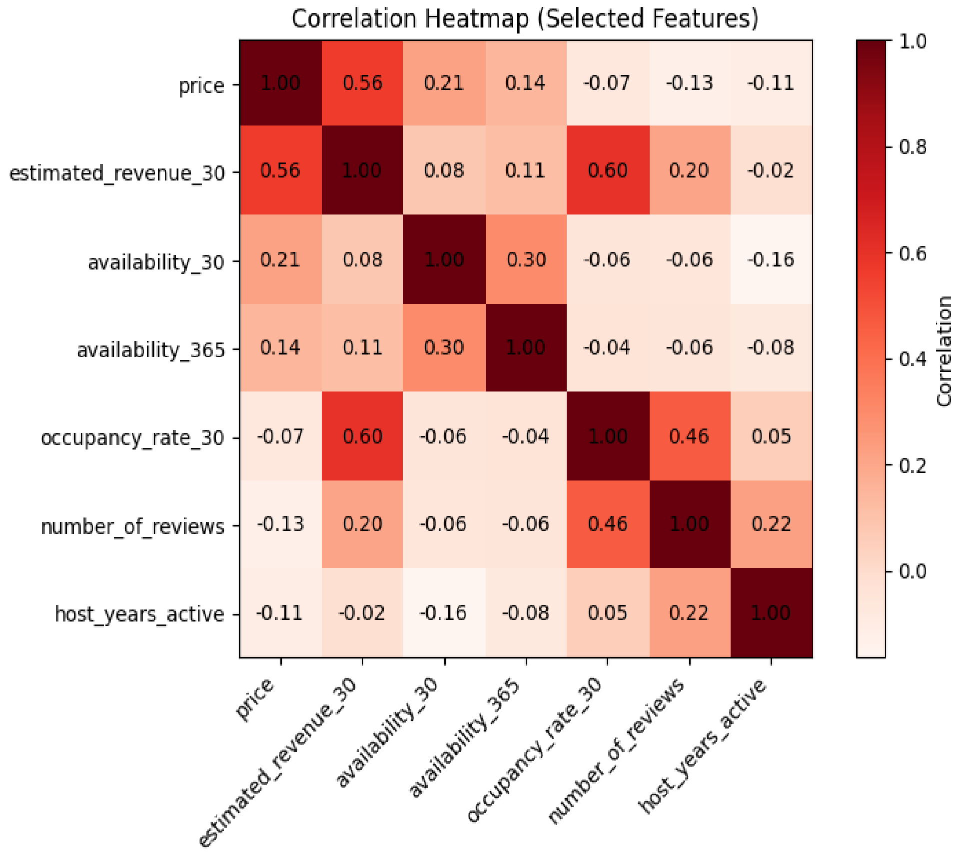
Availability_30

Shows distribution of how many nights are still free.



CORRELATION HEATMAP OF KEY NUMERIC FEATURES

- **Strong positive correlation**: occupancy_rate_30 with estimated_occupancy_30. estimated_revenue_30 with both price and occupancy.
- **Negative correlation:** availability_30 vs occupancy_rate_30 (more availability → fewer booked nights).
- **Moderate positive:** number_of_reviews vs occupancy (more reviews generally means higher occupancy).



“Correlation guided which variables to keep and which groups behave similarly.”

CLASSIFICATION SETUP (VERY HIGH VS REST)

- Continuous target: occupancy_rate_30 (0–1).
- We turned it into a binary classification:
  - Used quartiles (Q1–Q4).
  - Top quartile (Q4) = “Very High Occupancy” (label 1).
  - Q1–Q3 = “Others” (label 0).
- Train/test split:
  - 80% training, 20% testing.
  - Stratified by occupancy to keep balance.
- Preprocessing:
  - Numeric features scaled with StandardScaler.
  - Categorical already one-hot encoded in previous step.

SUPERVISED LEARNING MODELS

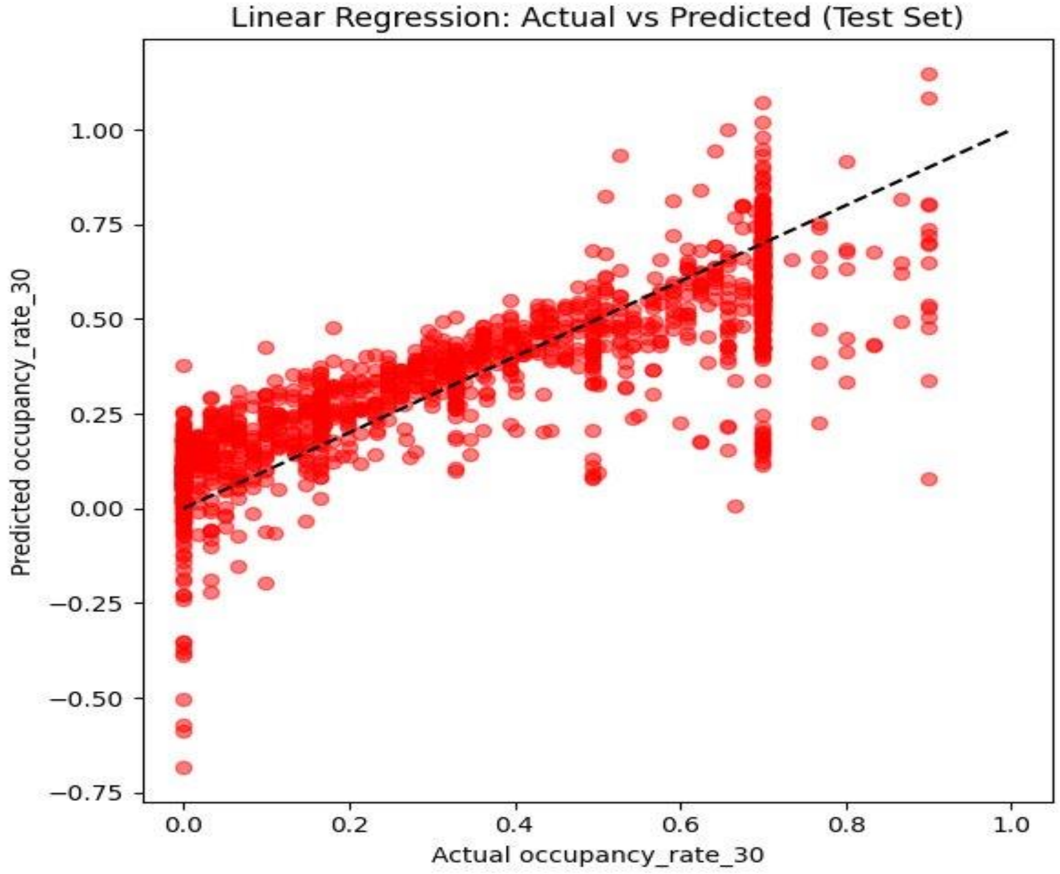
- Linear Regression
- Logistic Regression
- Random Forest
- Decision Tree
- Naive Bayes

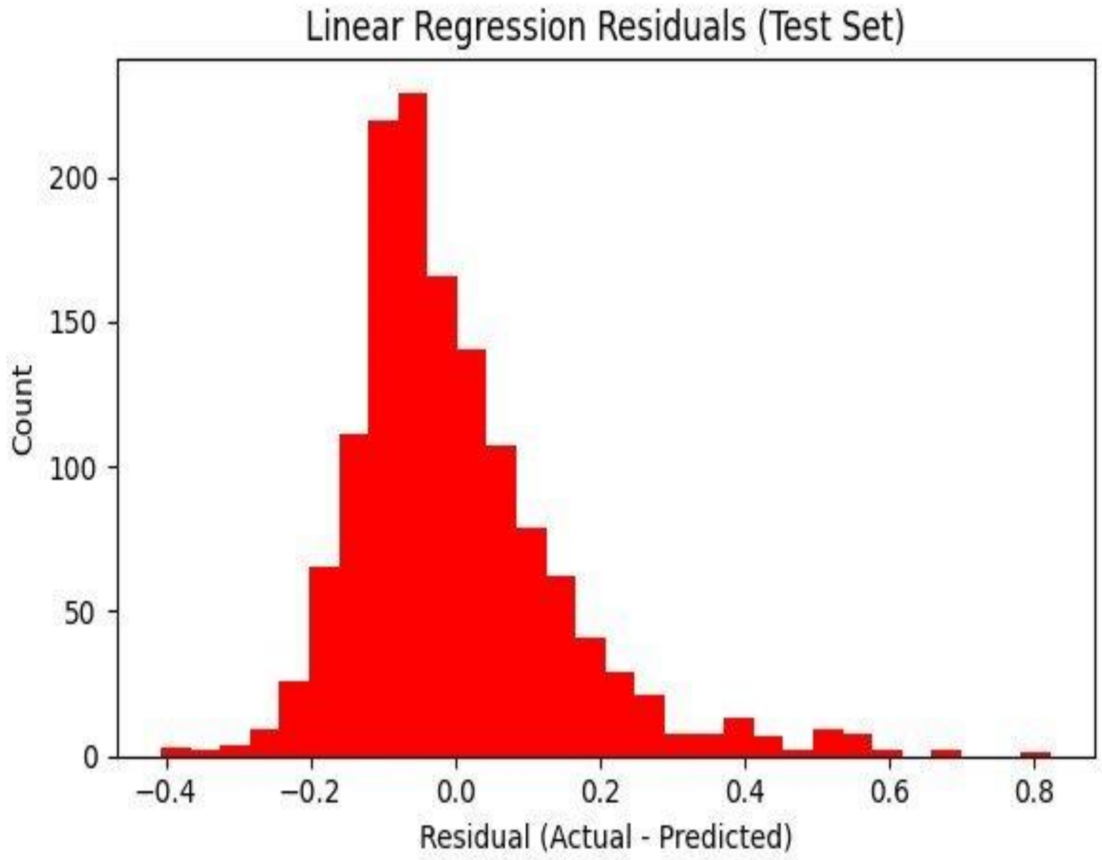
LINEAR REGRESSION

Target Variable: Occupancy_rate_30
Linear regression gives a baseline continuous prediction of occupancy.
It captures overall linear trends, but might miss non-linear interactions (e.g., combinations of price and availability).

Train r2	Test r2	Test MAE	Test RMSE
0.6795	0.6927	0.1088	0.1475

LINEAR REGRESSION





We mainly used it as a baseline and for comparison with more flexible models (Random Forest, Decision Tree).

LOGISTIC REGRESSION- THRESHOLDS 0.1–0.9

Logistic Regression fitted on the same binary target. Instead of using the default 0.5: Tested thresholds from 0.1 to 0.9.

For each threshold:

- Computed TP, FP, TN, FN, Accuracy.
- Later also Precision, Recall, F1.

Low threshold (0.1):

- Classifies more listings as “Very High” → high recall, low precision.

High threshold (0.9):

- Very strict → high precision, low recall.

Aim: find balanced cut-off where we don’t miss too many truly high-occupancy listings, but don’t over-label others as high.

THRESHOLD EVALUATION (PRECISION, RECALL, F1)

Threshold	TP	FP	TN	FN	Precision	Recall	F1_Score
0.1	303	198	840	32	0.6048	0.9045	0.7249
0.2	301	132	906	34	0.6952	0.8985	0.7839
0.3	293	85	953	42	0.7751	0.8746	0.8219
0.4	287	60	978	48	0.8271	0.8567	0.8416
0.5	269	46	992	66	0.8540	0.8030	0.8277
0.6	252	25	1013	83	0.9097	0.7522	0.8235
0.7	231	14	1024	104	0.9429	0.6896	0.7966
0.8	189	3	1035	146	0.9844	0.5642	0.7173
0.9	134	0	1038	201	1.0000	0.4000	0.5714

YOUDEN'S J: ANOTHER WAY TO FIND THE BEST CUTOFF

Threshold	TP	FP	TN	FN	Sensitivity_Recall	Specificity	J_stat
0.1	303	198	840	32	0.9045	0.8092	0.7137
0.2	301	132	906	34	0.8985	0.8728	0.7713
0.3	293	85	953	42	0.8746	0.9181	0.7927
0.4	**287**	**60**	**978**	**48**	**0.8567**	**0.9422**	**0.7989**
0.5	269	46	992	66	0.8030	0.9557	0.7587
0.6	252	25	1013	83	0.7522	0.9759	0.7282
0.7	231	14	1024	104	0.6896	0.9865	0.6761
0.8	189	3	1035	146	0.5642	0.9971	0.5613
0.9	134	0	1038	201	0.4000	1.0000	0.4000

LOGISTIC REGRESSION (CUTOFF = 0.4)

Input: all selected features after preprocessing.
Output: probability of Very High Occupancy.
Strengths: Simple, interpretable in terms of feature effects (signs).
Limitations: Assumes mostly linear relationships.

True Positive	False Positive	True Negative	False Negative
287	60	978	48

Accuracy	Precision	Recall	F1-Score
0.9213	0.8271	0.8567	0.8416

RANDOM FOREST CLASSIFIER

Can model non-linear interactions among features.
Often strong predictive performance, but harder to directly interpret.
Random Forest did well, but Decision Tree gave a better trade-off at this cutoff + more interpretability.

True Positive	False Positive	True Negative	False Negative
301	47	991	34

Accuracy	Precision	Recall	F1-score
0.9410	0.8649	0.8985	0.8814

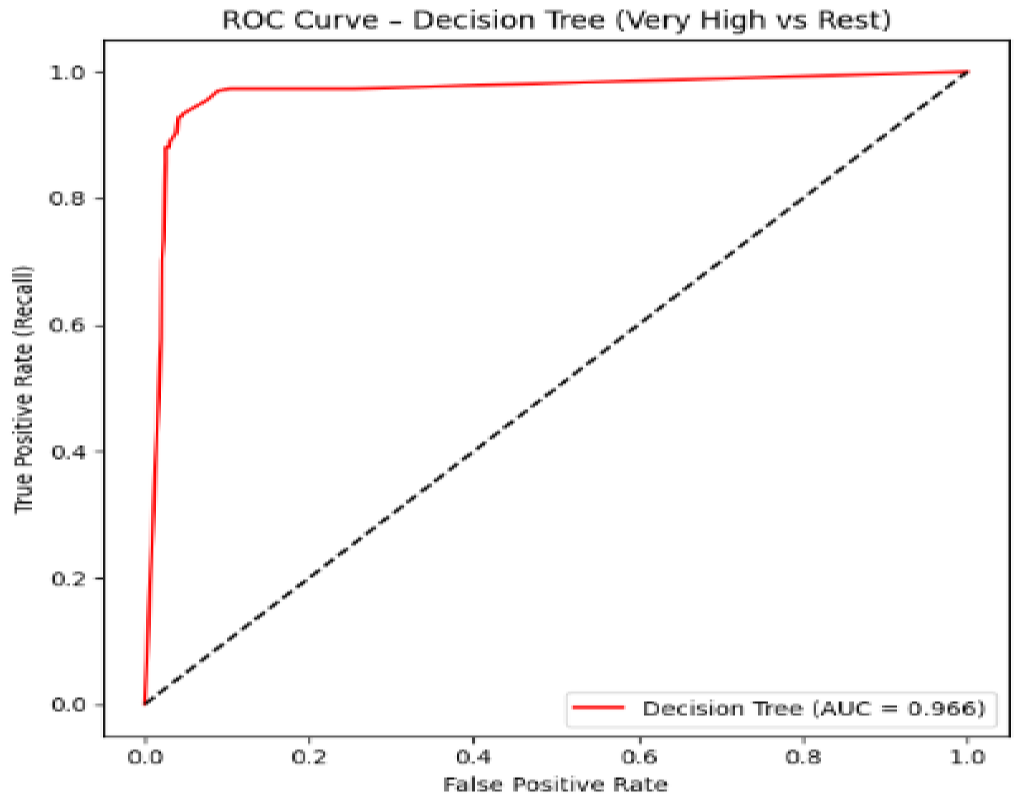
DECISION TREE

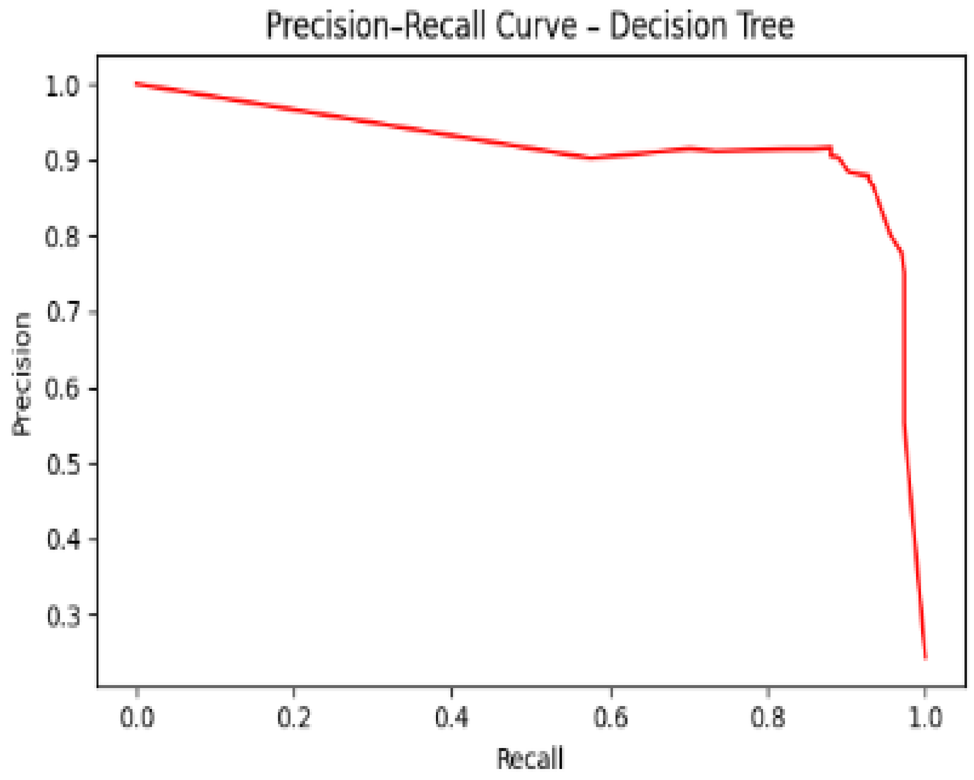
The model correctly separates most high-occupancy listings and provides simple, interpretable rules for hosts.

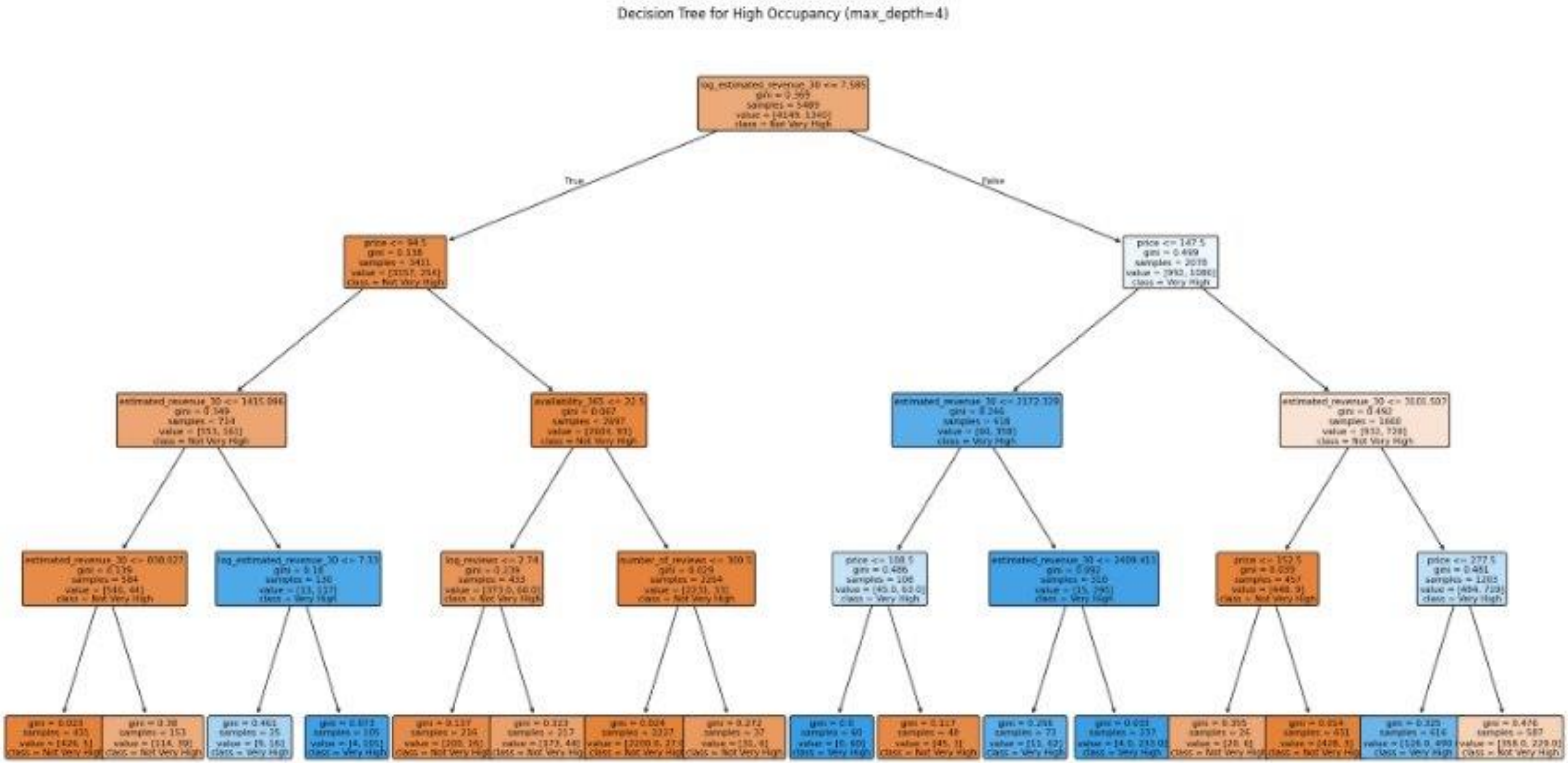
True Positive	False Positive	True Negative	False Negative
302	39	999	33

Accuracy	Precision	Recall	F1-score
0.9476	0.8856	0.9015	0.8935

Captures non-linear booking patterns while remaining easy to interpret for host-friendly insights.







Decision Tree for High Occupancy (max_depth=4)
True
False

DECISION TREE

Estimated_revenue_30 < = 1967.4?

Price < = 94.5?

Price < = 147.5?

If log_estimated_revenue>7.26 and estimated_revenue_30 < = 1524: More likely very high occupancy (class1)

If price is moderate and revenue is in certain band: More likely very high occupancy.
Very high price at these revenue levels can reduce probability of high occupancy.

NAIVE BAYES

Assumes features are **independent** given the class label.
Very fast and simple baseline.
At 0.4: Show key metrics (Accuracy, F1).
In our case:

- Performed **worse** than tree-based models.
- Still useful as a comparison baseline.

True Positive	False Positive	True Negative	False Negative
315	803	235	20

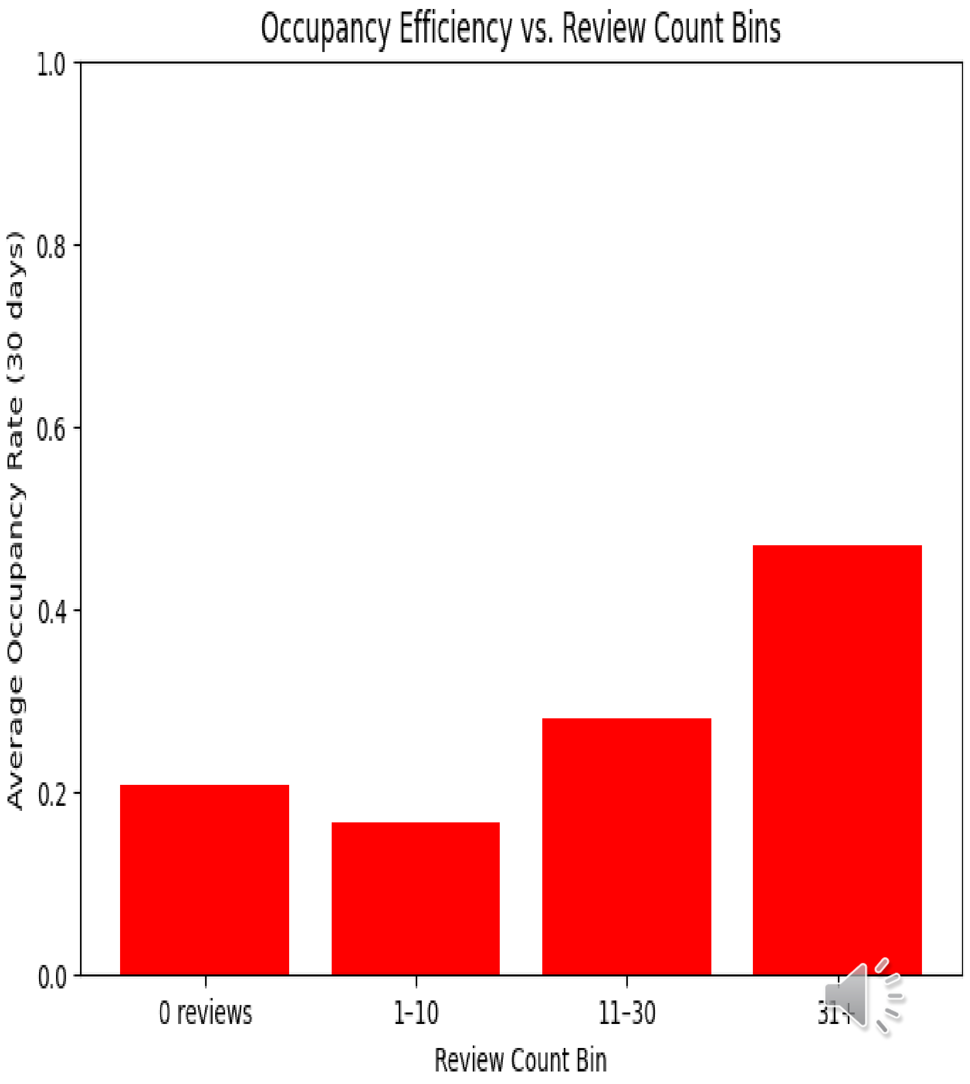
Accuracy	Precision	Recall	F1-Score
0.4006	0.2818	0.9403	0.4336

MODEL PERFORMANCE

Model	Cutoff	TP	FP	TN	FN	Accuracy	Precision	Recall	F1
Logistic Regression	0.4	287	60	978	48	0.9213	0.8271	0.8567	0.8416
Random Forest	0.4	301	47	991	34	0.9410	0.8649	0.8985	0.8814
Decision Tree	0.4	302	39	999	33	0.9476	0.8856	0.9015	0.8935
Naive Bayes	0.4	315	803	235	20	0.4006	0.2818	0.9403	0.4336

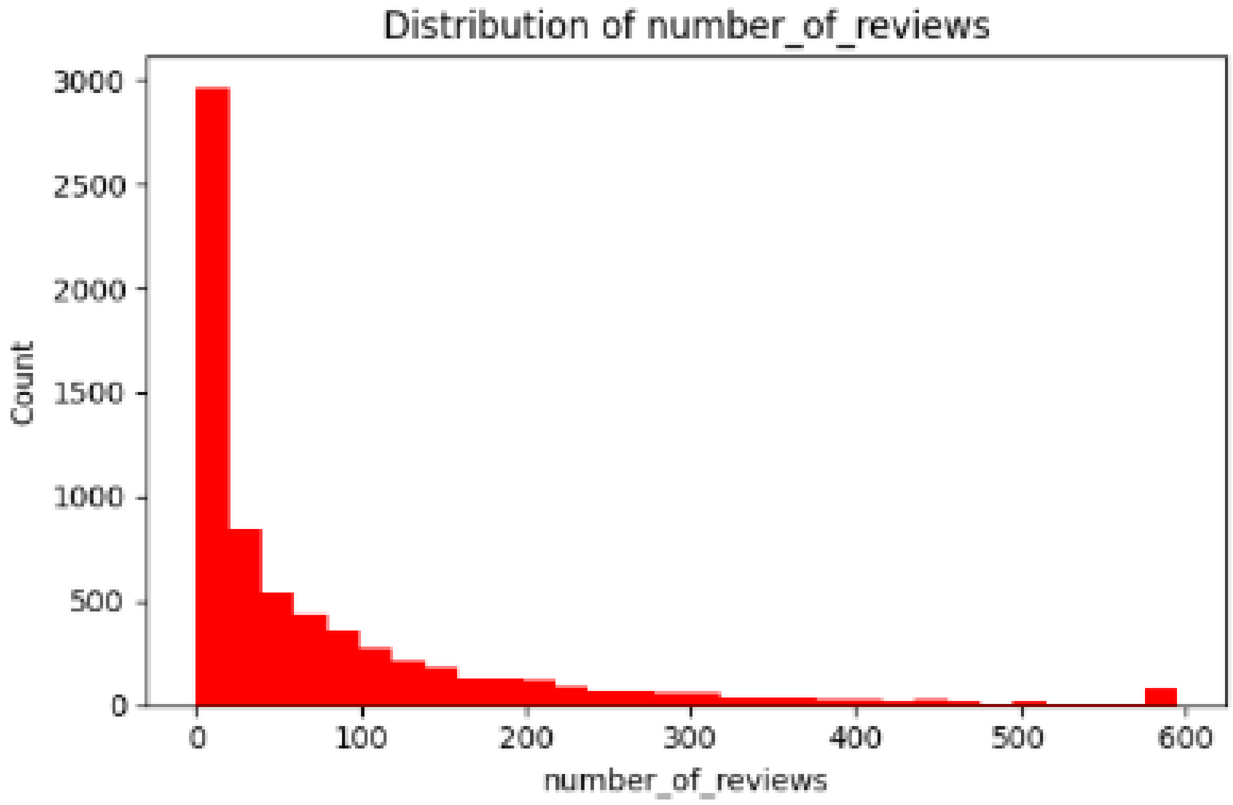
DO REVIEWS AFFECT OCCUPANCY?

- Yes – reviews matter, especially the number of reviews.
  - Listings with more reviews tend to have higher 30-day occupancy.
- Strength of effects:
  - review_count: strongest relationship with occupancy.
  - review_timespan_days: moderate – listings with reviews over a longer period tend to stay active and booked.
  - avg_review_length: weak – length of text isn't as important as having reviews at all.
- Practical host advice:
  - Encourage guests to leave reviews.
  - Maintain good service to get reviews consistently over time.
  - Focus on volume and recency of reviews more than long paragraphs.



Number_of_reviews

Very skewed: many listings with few reviews, few listings with a lot



CONCLUSION

The Decision Tree is the best model for our project because it provides the strongest combination of Precision (0.8856) and Recall (0.9015), meaning it correctly identifies most high-occupancy listings while keeping false positives low — and it does so in a way that is easy to interpret and explain to hosts.

THANK YOU :)